

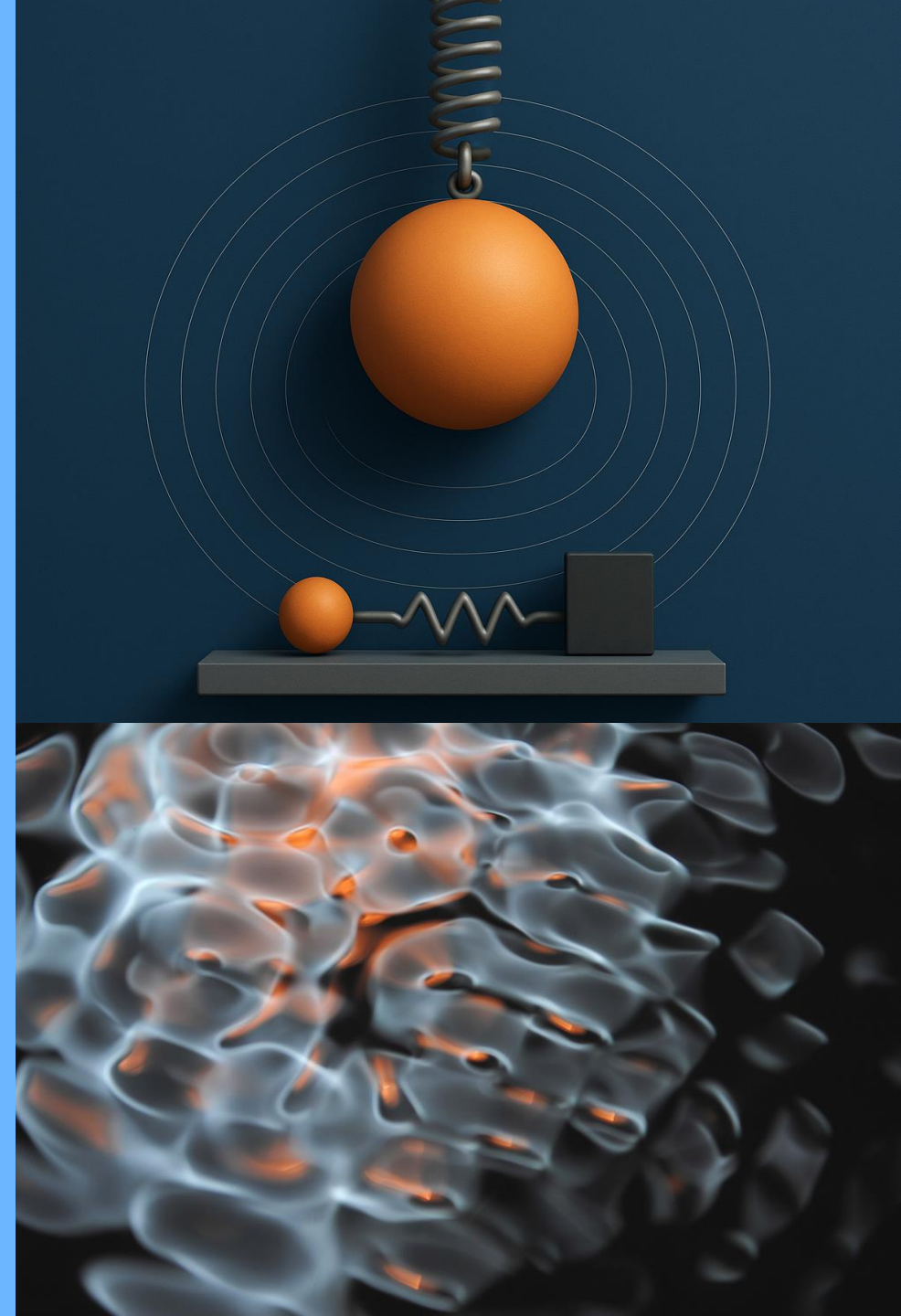


ES386

Dynamics of Vibrating Systems

Unit 6: Free multiple degree of freedom vibration, undamped lump model

UNIVERSITY
OF WARWICK





vevox

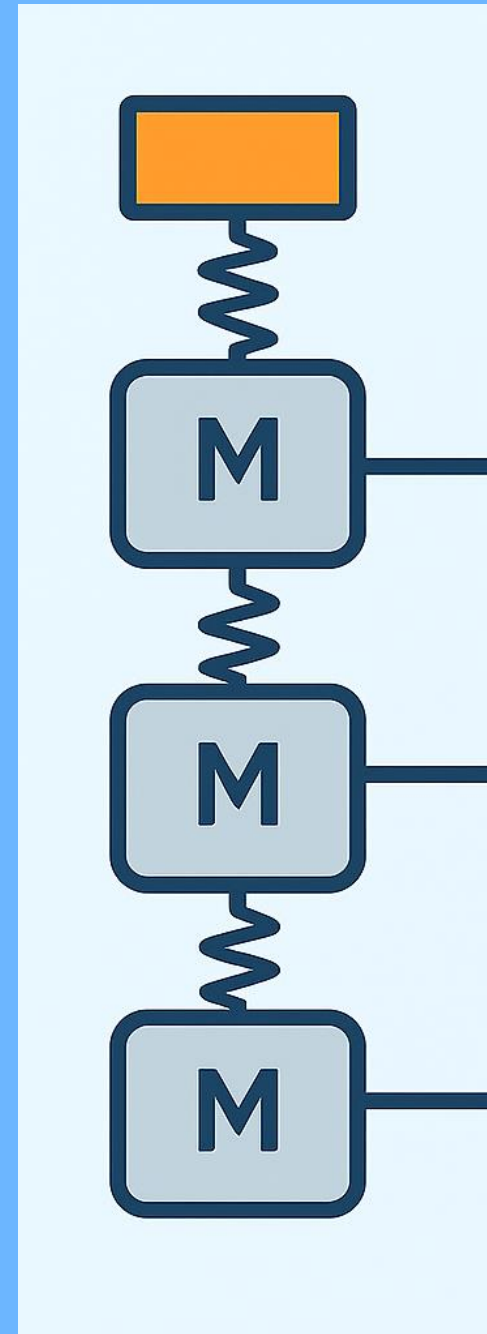
Audience Engagement





Plan for today

1. *Summary* of solving matrix equations
2. Visualising what they mean
3. Many degrees of freedom – easy matrix building for simple systems (aka heuristics)
4. Free-free systems



Free n-DoF vibrations:



How do you solve a problem like

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}$$



Solving equations of motion – deal with 2nd order ODE

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{K}\mathbf{x} = \mathbf{0}$$

The trial solution can be written as

$$x_i = A_i \cos(\omega t + \varphi)$$

Then,

$$\ddot{x}_i = -\omega^2 A_i \cos(\omega t + \varphi)$$

Rao 6.9

$$-\omega^2 \mathbf{M}\mathbf{x} + \mathbf{K}\mathbf{x} = \mathbf{0}$$

$$-\lambda \mathbf{M}\mathbf{x} + \mathbf{K}\mathbf{x} = \mathbf{0}$$

$$\mathbf{K}\mathbf{x} - \lambda \mathbf{M}\mathbf{x} = \mathbf{0}$$

$$\mathbf{M}^{-1}\mathbf{K}\mathbf{x} = \lambda \mathbf{x}$$

Final solution

$$x_i = \sum_j A_{ij} \cos(\omega_j t + \varphi_j)$$

i and j count degrees of freedom



Note on matrices and solution

Coupling types:

- **Static coupling:** K not diagonal (like translational motion of masses with springs)
- **Dynamic coupling:** M not diagonal (e.g. trolley-pendulum)

$$\lambda = \omega^2$$

$$K\mathbf{x} - \lambda M\mathbf{x} = \mathbf{0}$$

$$(M^{-1}K - \lambda I)X = 0$$

$$-\omega^2 M\mathbf{x} + K\mathbf{x} = \mathbf{0}$$

$$\left(\frac{1}{\omega^2}K - M\right)X = 0$$

$$\mu = \frac{1}{\omega^2}; \quad D = K^{-1}M$$

$$(\mu I - D)X = 0$$

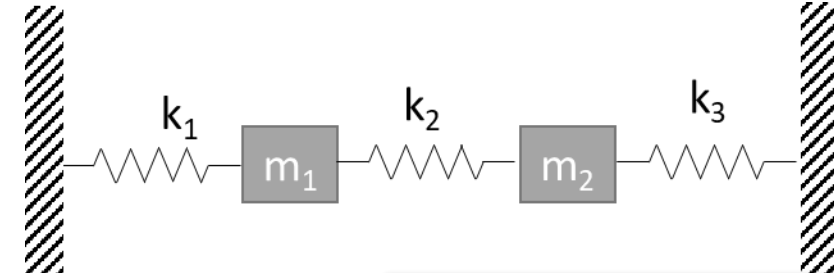


We can do it in Matlab!

$$-\omega^2 M \mathbf{x} + K \mathbf{x} = 0$$

Matlab can solve equations of the type

$$KX - \lambda MX = 0$$



The code is `[eigvector,eigvalue] = eig (K, M)` – remember to get ω back from substitution

For $m_1 = m_2 = 1.0\text{kg}$, all $k = 1.0\text{N/m}$:

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$K = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

```
>> M=[1 0; 0 1]

M =

     1     0
     0     1

>> K=[2 -1; -1 2]

K =

     2     -1
    -1     2

>> [eigvect, eigvalue]=eig(K,M)

eigvect =

   -0.7071   -0.7071
   -0.7071    0.7071

eigvalue =

     1     0
     0     3
```

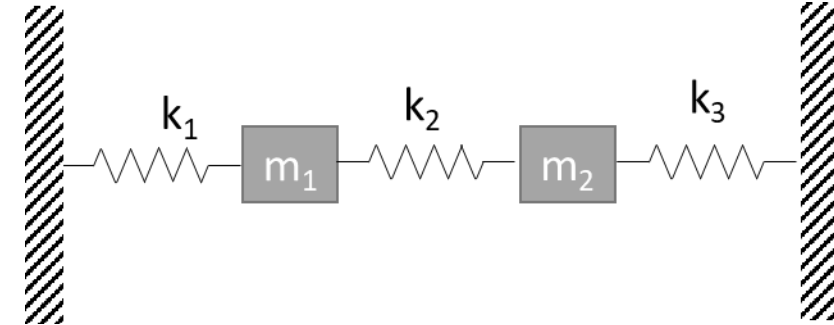


Matlab solution – what it means

$$-\omega^2 M \mathbf{x} + K \mathbf{x} = 0$$

Matlab can solve equations of the type

$$KX - \lambda MX = 0$$



The code is `[eigvector,eigvalue] = eig (K, M)` – remember to get ω back from substitution

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$
$$K = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

For $m_1 = m_2 = 1.0\text{kg}$, all $k = 1.0\text{N/m}$:

$$\omega^2 \begin{matrix} \text{eigvalues} = \\ \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix} \end{matrix}$$

$$\begin{bmatrix} A_1^{(1,2)} \\ A_2^{(1,2)} \end{bmatrix} \begin{matrix} \text{eigvectors} = \\ \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \end{matrix}$$

$$\omega_1 = 1 \text{ rad/s and } \omega_2 = 1.7 \text{ rad/s}$$

```
>> [eigvect, eigvalue]=eig(K,M)

eigvect =

   -0.7071   -0.7071
   -0.7071    0.7071

eigvalue =

     1     0
     0     3
```


Eigenvalue defines mode frequency

Eigenvector defines mode shape

2 inertia system demo – spring and carts

eigval =

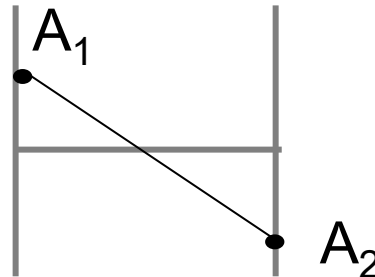
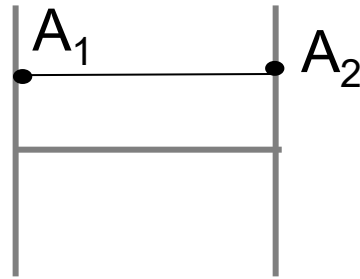
1 0
0 3

$\omega =$

1.0000 0
0 1.7321

eigvect =

1.0000 1.0000
1.0000 -1.0000



$$\begin{Bmatrix} x \\ x \end{Bmatrix} = \begin{Bmatrix} x_1(t) \\ x_2(t) \end{Bmatrix} = C_1 \begin{Bmatrix} A_1^{(1)} \\ A_2^{(1)} \end{Bmatrix} \cos(\omega_1 t - \varphi_1) + C_2 \begin{Bmatrix} A_1^{(2)} \\ A_2^{(2)} \end{Bmatrix} \cos(\omega_2 t - \varphi_2)$$

2-inertia, free-free

Start with very simple example

2 disks on a shaft

Twist in opposite directions,
then let go:

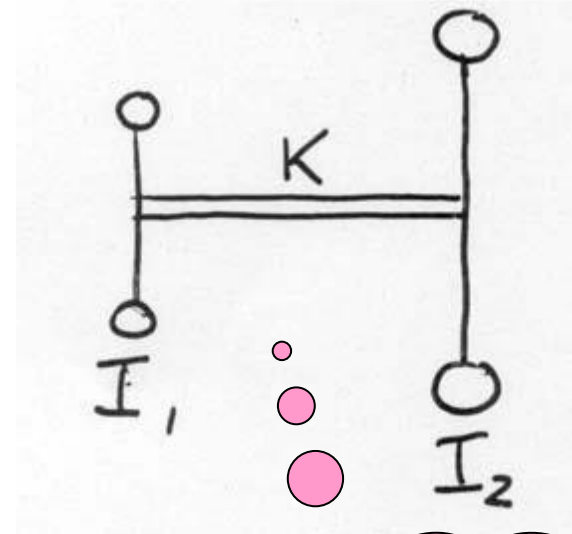
Isolated – momentum (H) conserved;

disks move opposite ways;

They come to origin at the same times;

One natural frequency, one node.

(Second mode frequency = 0)



2 inertias, so
where's the
2nd mode?

3 inertia system demo – rope!

For $m_1 = m_2 = m_3 = 1.0\text{kg}$, all $k = 1.0\text{N/m}$:

eigval =

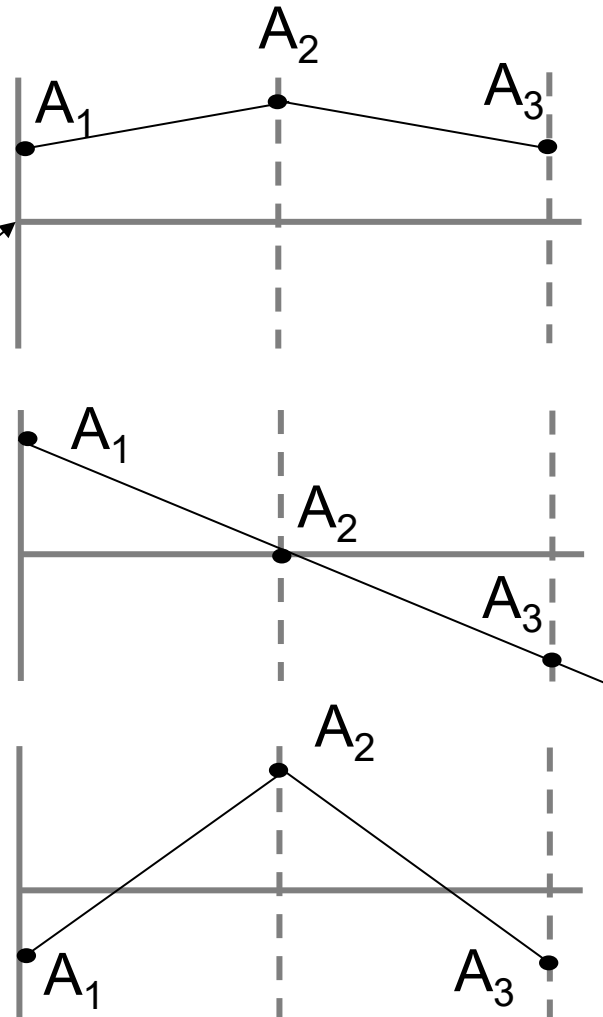
0.5858	0	0
0	2.0000	0
0	0	3.4142

$\omega =$

0.7654
1.4142
1.8478

eigvect =

0.7071	-1.0000	-0.7071
1.0000	0.0000	1.0000
0.7071	1.0000	-0.7071



Coupled Air Carts



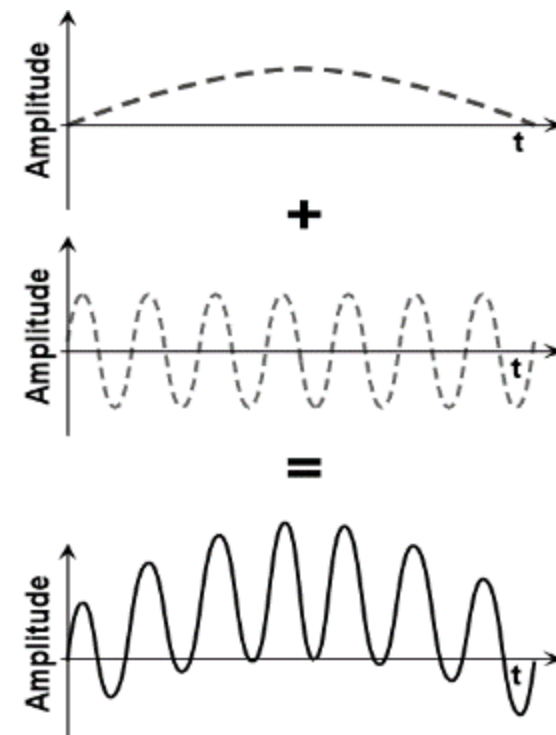


Physical intuitions

- ▶ Number of modes = Degrees of Freedom
- ▶ Finite number of degrees of freedom means a finite number of basic stable patterns of motion.
 - But many forms of transient motion are possible.
- ▶ A stable motion requires all inertias have the same oscillatory frequency.
- ▶ If all inertias have the same stable frequency, then they must move in (anti)phase or the spring forces change their balance and the motion will alter.
- ▶ For similar amplitudes, need more energy to create a mode with more nodes
- ▶ At most, one node between two adjacent inertias (or inertia and wall)
 - With n modes, $n-1$ nodes

Superposition principle

- ▶ Stable patterns happen only at certain **natural frequencies**.
- ▶ If system is (modelled as) **linear**, then these **simple patterns** can be **superposed** to show other patterns.
- ▶ Natural frequency and mode shapes depend only on the particular configuration of **stiffness and inertias**.
 - Independent of initial conditions
 - Within reason – too much energy and it breaks (linearity breaks down).



2 inertia system - full solution

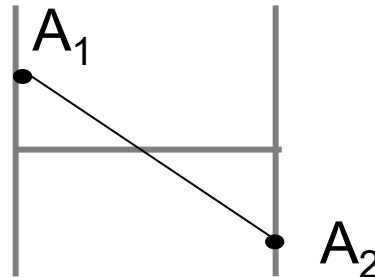
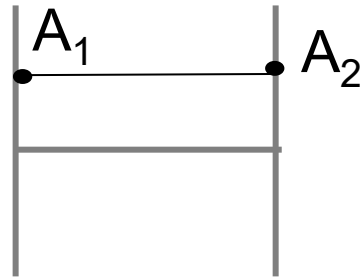
eigval =

$$\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}$$

$\omega =$

$$\begin{pmatrix} 1.0000 & 0 \\ 0 & 1.7321 \end{pmatrix}$$

eigvect =

$$\begin{pmatrix} 1.0000 & 1.0000 \\ 1.0000 & -1.0000 \end{pmatrix}$$


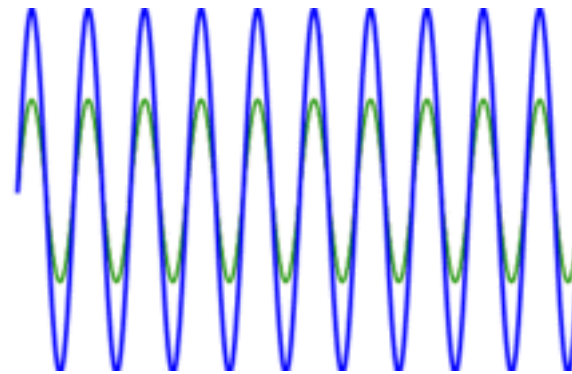
$$\begin{Bmatrix} x \\ x \end{Bmatrix} = \begin{Bmatrix} x_1(t) \\ x_2(t) \end{Bmatrix} = C_1 \begin{Bmatrix} A_1^{(1)} \\ A_2^{(1)} \end{Bmatrix} \cos(\omega_1 t - \varphi_1) + C_2 \begin{Bmatrix} A_1^{(2)} \\ A_2^{(2)} \end{Bmatrix} \cos(\omega_2 t - \varphi_2)$$

Mode superposition – what if $\omega_1 \approx \omega_2$

- ▶ The sum of two cosines can be rewritten as product.

$$\cos \omega_1 t + \cos \omega_2 t = 2 \cos \frac{\omega_1 + \omega_2}{2} \cos \frac{\omega_1 - \omega_2}{2}$$

<http://youtu.be/RoSYKPTdlxs>

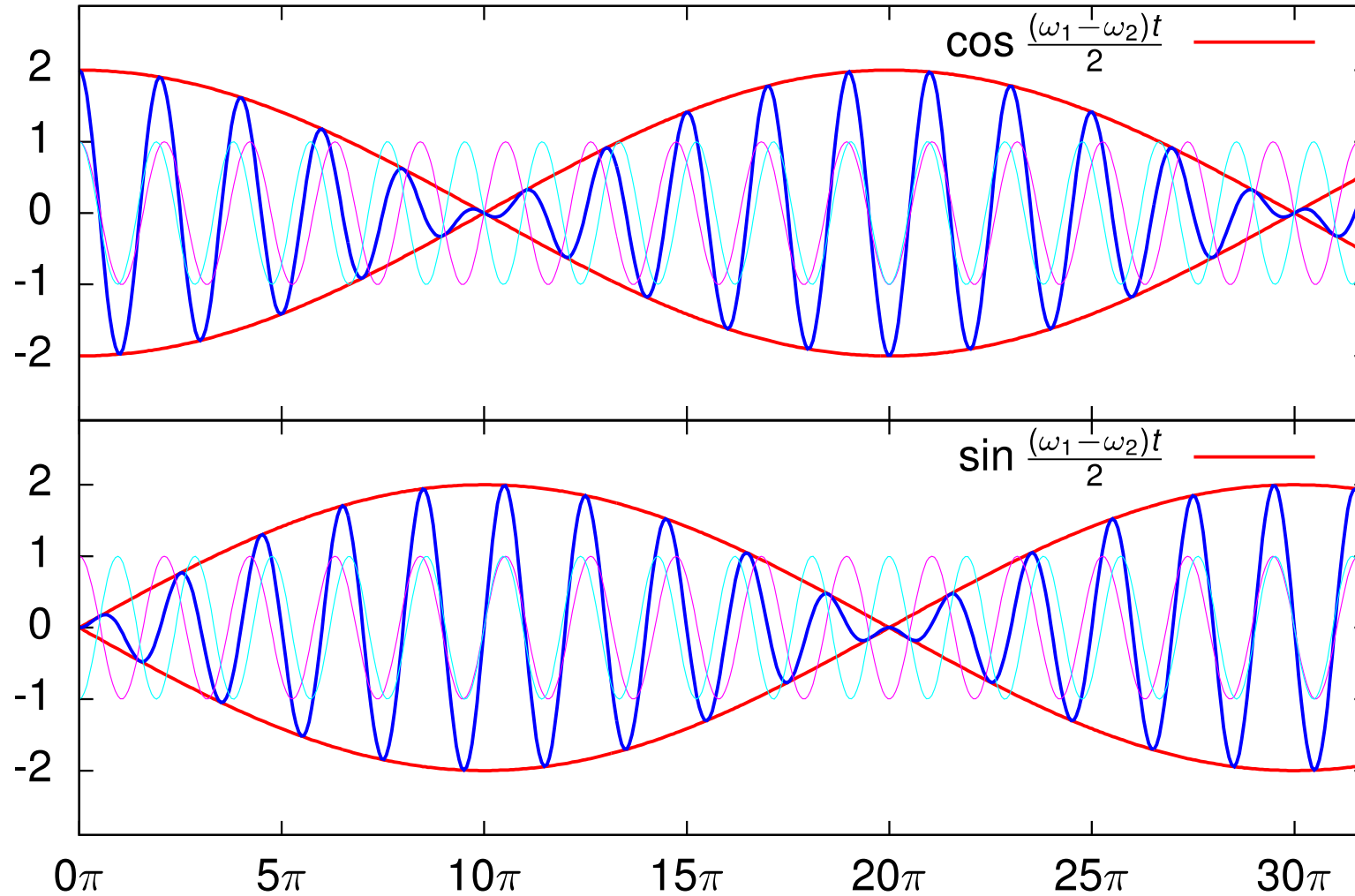


Starting from
 $\omega_1 = \omega_2$, increase
difference up to 25%

<http://goo.gl/zCcteh>

By Adjwilley [CC BY-SA 3.0](#)

Mode superposition - beat



$$\omega_{1,2} = (1 \pm 0.05) \text{ rad/s}$$



Summary

- ▶ Free vibration in modes
- ▶ A mode: all inertias same frequency, same phase, fixed amplitude ratios
- ▶ Some modes have nodes
- ▶ Generally we get a superposition of normal modes
- ▶ Superposition: Part of mode 1, part of mode 2, phase
- ▶ Initial conditions: positions & velocities
 $x_1(t_0), \dot{x}_1(t_0), x_2(t_0), \dot{x}_2(t_0)$
determine how much of each mode

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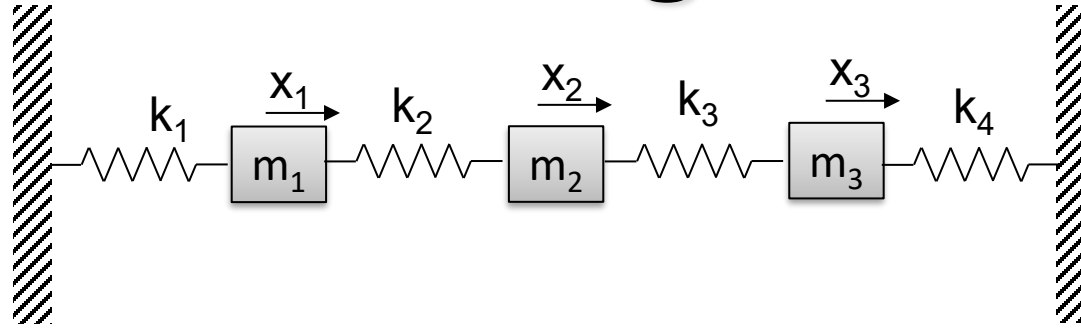
Heuristics

Heuristics are the strategies derived from previous experiences with similar problems

Patterns emerging



Free Axial Vibration – general case setup



3 DoF is enough to show the general patterns for n-DoF.

Energy terms:

$$T = \frac{1}{2}m_1\dot{x}_1^2 + \frac{1}{2}m_2\dot{x}_2^2 + \frac{1}{2}m_3\dot{x}_3^2$$

$$V = \frac{1}{2}k_1x_1^2 + \frac{1}{2}k_2(x_1 - x_2)^2 + \frac{1}{2}k_3(x_2 - x_3)^2 + \frac{1}{2}k_4x_3^2$$

$$T = \sum_{i=1}^n \frac{1}{2}m_i\dot{x}_i^2$$

$$V = \sum_{i=1}^n \frac{1}{2}k_i(x_i - x_{i-1})^2$$

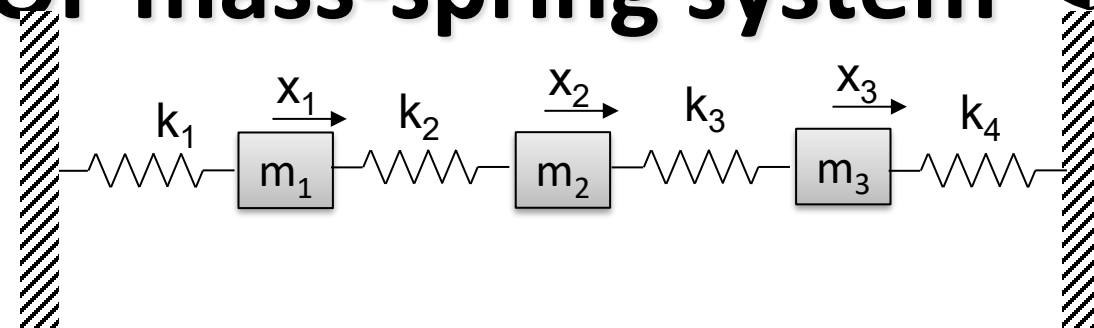
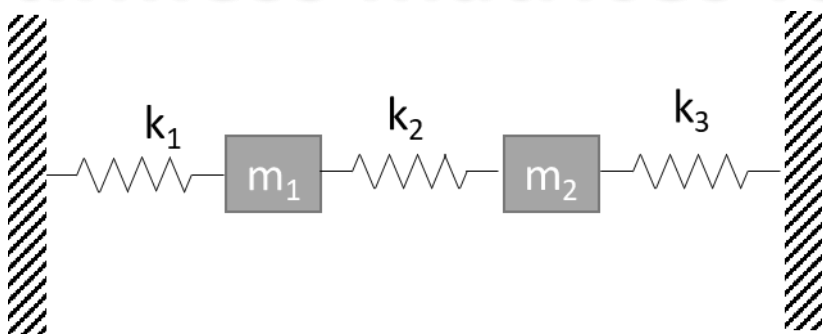
Free axial vibration – matrix form

For 3-DoF, the full matrix form is

$$\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{pmatrix} + \begin{bmatrix} (k_1 + k_2) & -k_2 & 0 \\ -k_2 & (k_2 + k_3) & -k_3 \\ 0 & -k_3 & (k_3 + k_4) \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$



Stiffness matrices for multi-DOF mass-spring system



$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 + k_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} \begin{pmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{pmatrix} + \begin{bmatrix} (k_1 + k_2) & -k_2 & 0 \\ -k_2 & (k_2 + k_3) & -k_3 \\ 0 & -k_3 & (k_3 + k_4) \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0$$

- ▶ Diagonal element K_{ii} : the sum of stiffnesses of all springs connected to M_i
- ▶ Other elements K_{ij} : negative stiffness of springs directly connected between M_i and M_j . If there are two or more springs, take the equivalent element.



Same rules work for damping matrices!

- ▶ Diagonal element C_{ii}
 - the sum of damping coefficients of all damping elements connected to M_i
- ▶ Other element C_{ij}
 - negative damping coefficients of dampers directly connected between M_i and M_j . If there are two or more dampers, take the equivalent element.
 - We will build some on Friday!



Summary 3 DoF

From patterns for 3 DoF axial system we derived

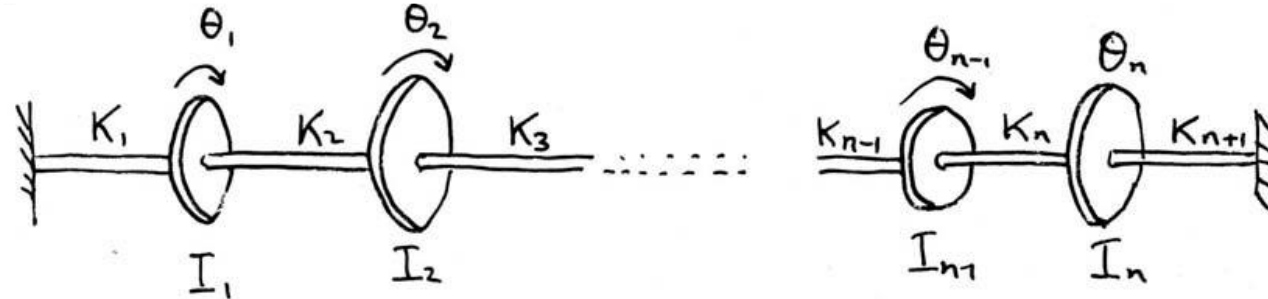
- ▶ rules for writing out energy terms
- ▶ rules for m , k and c matrices
- ▶ Same works for torsional motion



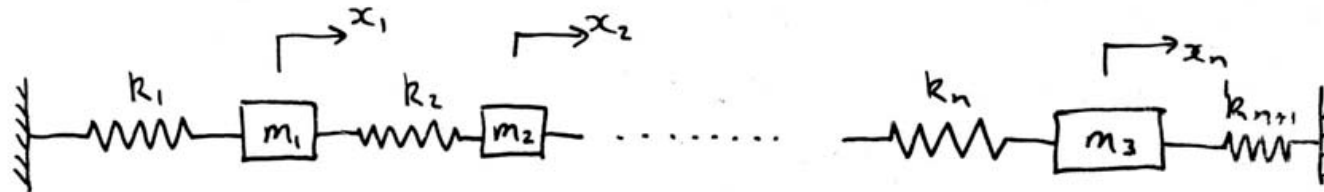
Torsional systems

Torsional Vibration – Mode Shapes

The torsional lumped system:



Corresponds exactly to the axial one



The generic energy terms are

$$T = \sum_{i=1}^n \frac{1}{2} I_i \dot{\theta}_i^2$$

$$V = \sum_{i=1}^n \frac{1}{2} K_i (\theta_i - \theta_{i-1})^2$$



Summary: free n-DoF vibrations

- ▶ Systems with n degrees of freedom have n modes
- ▶ Modes characterised by frequency and amplitude ratios (modal shape)
- ▶ We can use heuristic approach to build M and K and C matrices
- ▶ Lagrange needs to be used for complex systems (linked motions)



For the next class

- ▶ We will look at an equivalence problem in the light of Lagrange and heuristics, build matrices for more complex systems while looking at past exam papers and revisit equivalent damping + practice calculating number of DoF
- ▶ Watch the videos
- ▶ Only one example sheet this week, so for the challenge task, look through:
 - example 6.11 (Rao, p.629) – finding frequencies and mode shapes in 3DoF
 - 6.14 (Rao, p.641) - free-free translational system, 3 DoF
 - 6.15 (Rao p 644) - using initial conditions: for example 6.11, 3 DoF